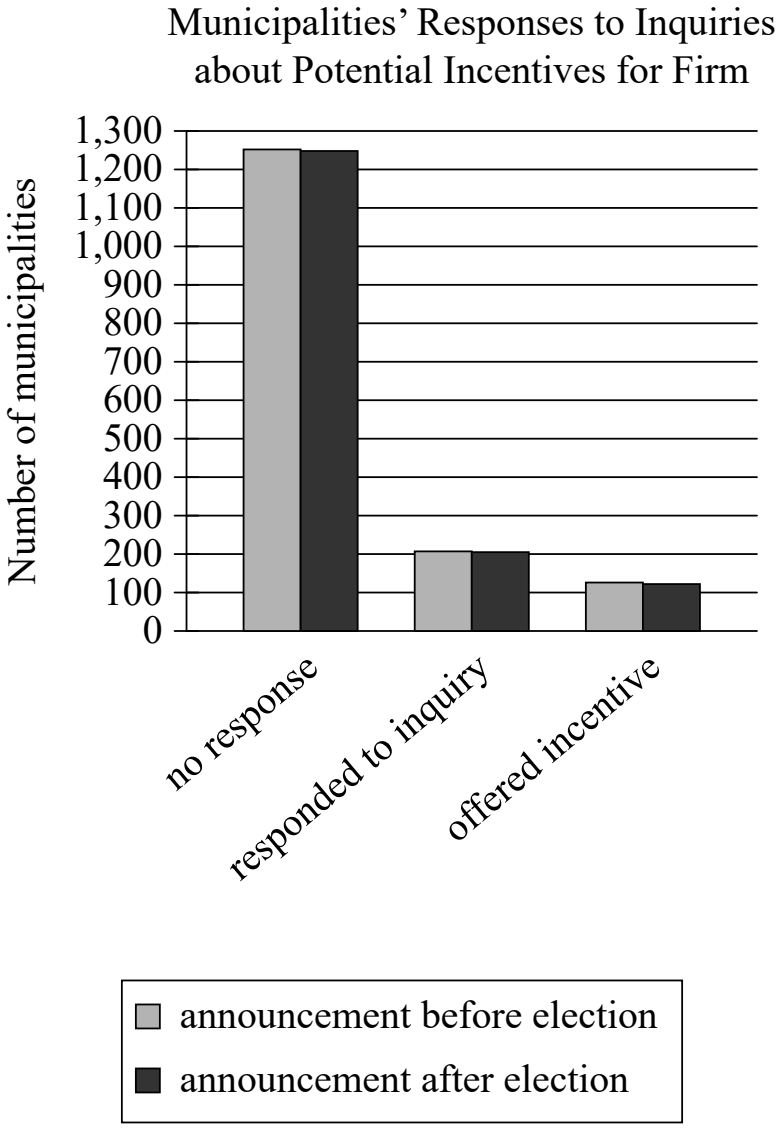
ID: a15b3219



In the United States, firms often seek incentives from municipal governments to expand to those municipalities. A team of political scientists hypothesized that municipalities are much more likely to respond to firms and offer incentives if expansions can be announced in time to benefit local elected officials than if they can't. The team contacted officials in thousands of municipalities, inquiring about incentives for a firm looking to expand and indicating that the firm would announce its expansion on a date either just before or just after the next election.

Which choice best describes data from the graph that weaken the team's hypothesis?

A. A large majority of the municipalities that received an inquiry mentioning plans for an announcement before the next election didn't respond to the inquiry.

B. The proportion of municipalities that responded to the inquiry or offered incentives didn't substantially differ across the announcement timing conditions.

C. Only around half the municipalities that responded to inquiries mentioning plans for an announcement before the next election offered incentives.

D. Of the municipalities that received an inquiry mentioning plans for an announcement date after the next election, more than 1,200 didn't respond and only around 100 offered incentives.

ID: 7a1877be

Nucleobase Concentrations from Murchison Meteorite and Soil Samples in Parts per Billion

Nucleobase	Murchison meteorite sample 1	Murchison meteorite sample 2	Murchison soil sample
Isoguanine	0.5	0.04	not detected
Purine	0.2	0.02	not detected
Xanthine	39	3	1
Adenine	15	1	40
Hypoxanthine	24	1	2

Employing high-performance liquid chromatography—a process that uses pressurized water to separate material into its component molecules—astrochemist Yashiro Oba and colleagues analyzed two samples of the Murchison meteorite that landed in Australia as well as soil from the landing zone of the meteorite to determine the concentrations of various organic molecules. By comparing the relative concentrations of types of molecules known as nucleobases in the Murchison meteorite with those in the soil, the team concluded that there is evidence that the nucleobases in the Murchison meteorite formed in space and are not the result of contamination on Earth.

Which choice best describes data from the table that support the team's conclusion?

A. Isoguanine and purine were detected in both meteorite samples but not in the soil sample.

B. Adenine and xanthine were detected in both of the meteorite samples and in the soil sample.

C. Hypoxanthine and purine were detected in both the Murchison meteorite sample 2 and in the soil sample.

D. Isoguanine and hypoxanthine were detected in the Murchison meteorite sample 1 but not in sample 2.

ID: 71904085

Linguist Deborah Tannen has cautioned against framing contentious issues in terms of two highly competitive perspectives, such as pro versus con. According to Tannen, this debate-driven approach can strip issues of their complexity and, when used in front of an audience, can be less informative than the presentation of multiple perspectives in a noncompetitive format. To test Tannen's hypothesis, students conducted a study in which they showed participants one of three different versions of local news commentary about the same issue. Each version featured a debate between two commentators with opposing views, a panel of three commentators with various views, or a single commentator.

Which finding from the students' study, if true, would most strongly support Tannen's hypothesis?

A. On average, participants perceived commentators in the debate as more knowledgeable about the issue than commentators in the panel.

B. On average, participants perceived commentators in the panel as more knowledgeable about the issue than the single commentator.

C. On average, participants who watched the panel correctly answered more questions about the issue than those who watched the debate or the single commentator did.

D. On average, participants who watched the single commentator correctly answered more questions about the issue than those who watched the debate did.

ID: 04cbeca3

In 1534 CE, King Henry VIII of England split with the Catholic Church and declared himself head of the Church of England, in part because Pope Clement VII refused to annul his marriage to Catherine of Aragon. Two years later, Henry VIII introduced a policy titled the Dissolution of the Monasteries that by 1540 had resulted in the closure of all Catholic monasteries in England and the confiscation of their estates. Some historians assert that the enactment of the policy was primarily motivated by perceived financial opportunities.

Which quotation from a scholarly article best supports the assertion of the historians mentioned in the text?

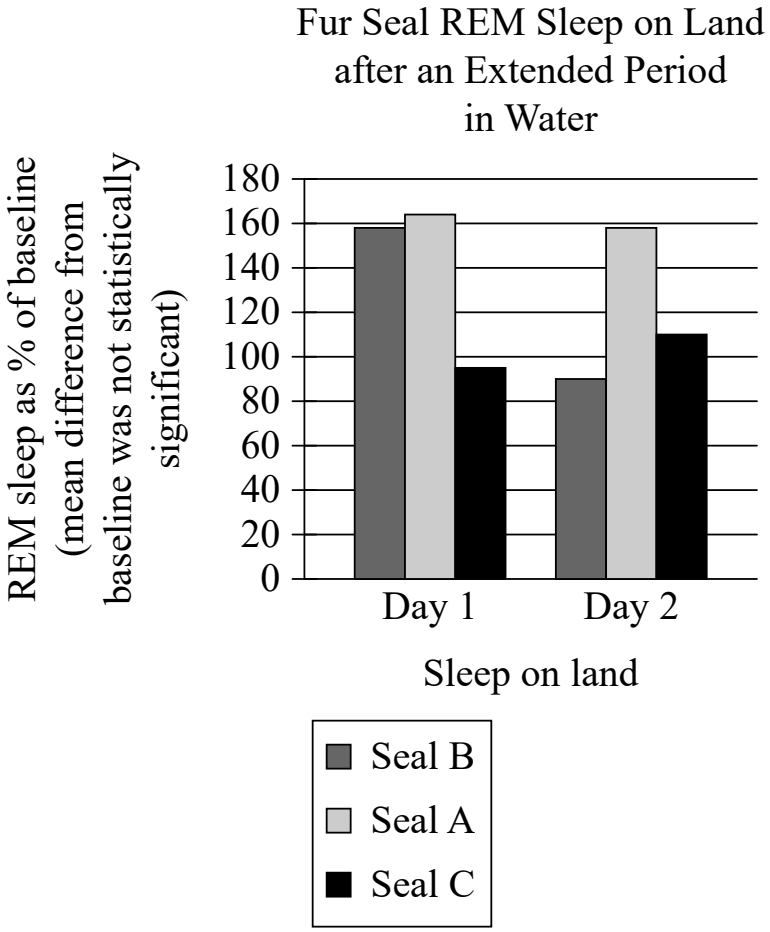
A. "At the time of the Dissolution of the Monasteries, about 2 percent of the adult male population of England were monks; by 1690, the proportion of the adult male population who were monks was less than 1 percent."

B. "A contemporary description of the Dissolution of the Monasteries, Michael Sherbrook's *Falle of the Religious Howses*, recounts witness testimony that monks were allowed to keep the contents of their cells and that the monastery timber was purchased by local yeomen."

C. "In 1535, the year before enacting the Dissolution of the Monasteries, Henry commissioned a survey of the value of church holdings in England—the work, performed by sheriffs, bishops, and magistrates, began that January and was swiftly completed by the summer."

D. "The October 1536 revolt known as the Pilgrimage of Grace had several economic motives: high food prices due to a poor harvest the prior year; the Dissolution of the Monasteries, which closed reliable sources of food and shelter for many; and rents and taxes throughout Northern England that were not merely high but predatory."

ID: ccb1ab92



Research suggests that REM sleep in animals is homeostatically regulated: animals compensate for periods of REM sleep deprivation by increasing subsequent REM sleep. When on land, fur seals get enough REM sleep, but during the weeks they're in the water, they get almost none. In a study of fur seals' sleep habits, researchers recorded the REM sleep (as a percentage of baseline) of fur seals once they had returned to land. They concluded that REM sleep may not be homeostatically regulated in fur seals, citing as evidence the fact that the seals in the study _____

Which choice most effectively uses data from the graph to complete the text?

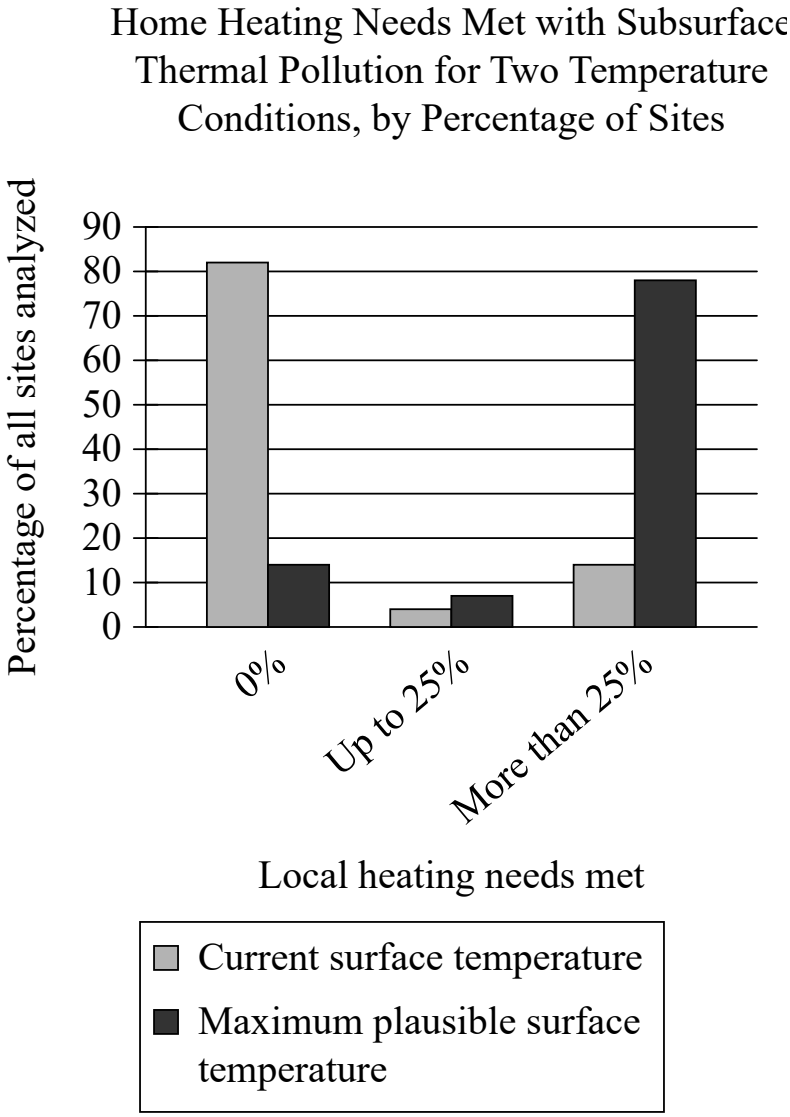
A. didn't show significantly less REM sleep during the second day after returning to land than they did during the first day.

B. showed no significant differences from one another in baseline levels of REM sleep.

C. didn't consistently demonstrate a significant increase in REM sleep after their period of deprivation in the water.

D. showed no significant difference between REM sleep after returning to land and REM sleep while in the water.

ID: be19faa1



Urbanization, industrialization, and the warming climate create thermal pollution (excess heat) in the shallow subsurface soil. Susanne A. Benz and colleagues analyzed thousands of sites on three continents under one scenario in which surface temperature remains at the current level and under another in which the surface reaches the maximum plausible temperature. They then categorized each site according to the percentage of local home heating needs that could be met using this excess subsurface heat. The team concluded that if surface temperature approaches the maximum plausible level, the percentage of sites where thermal pollution could feasibly contribute to meeting home heating needs will increase.

Which choice best describes data in the graph that support Benz and colleagues' conclusion?

A. Under both temperature conditions, less than 10% of sites were in the up-to-25% group, but at the maximum plausible surface temperature, almost 80% of sites could have all their local heating needs met by thermal pollution.

B. At current surface temperatures, more than 80% of the sites have no need for supplemental local home heating from subsurface thermal pollution, but at the maximum plausible surface temperature, more than 70% of sites exhibit significantly greater home heating needs.

C. At current surface temperatures, more than 80% of sites can meet, at most, 25% of local home heating needs with subsurface thermal pollution, but at the maximum plausible surface temperature, more than 80% of sites can meet greater than 25% of local home heating needs.

D. At current surface temperatures, more than 80% of the sites cannot use subsurface thermal pollution to meet any portion of local home heating needs, but at the maximum plausible surface temperature, that percentage drops below 20%.

ID: dd1757fd

Neural networks are computer models intended to reflect the organization of human brains and are often used in studies of brain function. According to an analysis of 11,000 such networks, Rylan Schaeffer and colleagues advise caution when drawing conclusions about brains from observations of neural networks. They found that when attempting to mimic grid cells (brain cells used in navigation), while 90% of the networks could accomplish navigation-related tasks, only about 10% of those exhibited any behaviors similar to those of grid cells. But even this approximation of grid-cell activity has less to do with similarity between the neural networks and biological brains than it does with the rules programmed into the networks.

Which finding, if true, would most directly support the claim in the underlined sentence?

A. The rules that allow for networks to exhibit behaviors like those of grid cells have no equivalent in the function of biological brains.

B. The networks that do not exhibit behaviors like those of grid cells were nonetheless programmed with rules that had proven useful in earlier neural-network studies.

C. Neural networks can often accomplish tasks that biological brains do, but they are typically programmed with rules to model multiple types of brain cells simultaneously.

D. Once a neural network is programmed, it is trained on certain tasks to see if it can independently arrive at processes that are similar to those performed by biological brains.

ID: 09f9edb0

In the 1980s, many musicians and journalists in the English-speaking world began to draw attention to music from around the globe—such as mbaqanga from South Africa and quan họ from Vietnam—that can't be easily categorized according to British or North American popular music genres, typically referring to such music as "world music." While some scholars have welcomed this development for bringing diverse musical forms to prominence in countries where they'd previously been overlooked, musicologist Su Zheng claims that the concept of world music homogenizes highly distinct traditions by reducing them all to a single category.

Which finding about mbaqanga and quan họ, if true, would most directly support Zheng's claim?

A. Mbaqanga and quan họ developed independently of each other and have little in common musically.

B. Mbaqanga is significantly more popular in the English-speaking world than quan họ is.

C. Mbaqanga and quan họ are now performed by a diverse array of musicians with no direct connections to South Africa or Vietnam.

D. Mbaqanga and quan họ are highly distinct from British and North American popular music genres but similar to each other.

ID: ec93e52c

Archaeologist Petra Vaiglova, anthropologist Xinyi Liu, and their colleagues investigated the domestication of farm animals in China during the Bronze Age (approximately 2000 to 1000 BCE). By analyzing the chemical composition of the bones of sheep, goats, and cattle from this era, the team determined that wild plants made up the bulk of sheep's and goats' diets, while the cattle's diet consisted largely of millet, a crop cultivated by humans. The team concluded that cattle were likely raised closer to human settlements, whereas sheep and goats were allowed to roam farther away.

Which finding, if true, would most strongly support the team's conclusion?

A. Analysis of the animal bones showed that the cattle's diet also consisted of wheat, which humans widely cultivated in China during the Bronze Age.

B. Further investigation of sheep and goat bones revealed that their diets consisted of small portions of millet as well.

C. Cattle's diets generally require larger amounts of food and a greater variety of nutrients than do sheep's and goats' diets.

D. The diets of sheep, goats, and cattle were found to vary based on what the farmers in each Bronze Age settlement could grow.

ID: 39e440e4

Archaeologists have held that the Casarabe culture, which emerged in the southwestern Amazon basin in the first millennium CE, was characterized by a sparse, widely distributed population and little intervention in the surrounding wilderness. Recently, however, archaeologist Heiko Prümers and colleagues conducted a study of the region using remote-sensing technology that enabled them to create three-dimensional images of the jungle-covered landscape from above, and the researchers concluded that the Casarabe people developed a form of urbanism in the Amazon basin.

Which finding about the remote-sensing images, if true, would most directly support Prümers and colleagues' conclusion?

A. They show shapes consistent with widely separated settlements of roughly equal small size surrounded by uncultivated jungle.

B. They show shapes consistent with long-distance footpaths running from Casarabe territories to large cities outside the region inhabited by the Casarabe people.

C. They show shapes consistent with scattered small farms created by clearing jungle areas near sources of fresh water.

D. They show shapes consistent with monumental platforms and dense central settlements linked to smaller settlements by a system of canals and roadways.

ID: 3f4ab688

In a research paper, a student criticizes some historians of modern African politics, claiming that they have evaluated Patrice Lumumba, the first prime minister of what is now the Democratic Republic of the Congo, primarily as a symbol rather than in terms of his actions.

Which quotation from a work by a historian would best illustrate the student's claim?

A. "Lumumba is a difficult figure to evaluate due to the starkly conflicting opinions he inspired during his life and continues to inspire today."

B. "The available information makes it clear that Lumumba's political beliefs and values were largely consistent throughout his career."

C. "Lumumba's practical accomplishments can be passed over quickly; it is mainly as the personification of Congolese independence that he warrants scholarly attention."

D. "Many questions remain about Lumumba's ultimate vision for an independent Congo; without new evidence coming to light, these questions are likely to remain unanswered."

ID: 156ff681

Many governments that regularly transfer money to individuals—to provide supplemental incomes for senior citizens, for example—have long done so electronically, but other countries typically have distributed physical money and have only recently developed electronic transfer infrastructure. Researchers studied the introduction of an electronic transfer system in one such location and found that recipients of electronic transfers consumed a different array of foods than recipients of physical transfers of the same amount did. One potential explanation for this result is that individuals conceive of and allocate funds in physical money differently than they conceive of and allocate funds in electronic form.

Which finding from the study, if true, would most directly weaken the potential explanation?

A. Recipients of electronic transfers typically spent their funds at a slower rate than recipients of physical transfers did.

B. Nearly every recipient of an electronic transfer withdrew the entire amount in physical money shortly after receiving the transfer.

C. Recipients of physical transfers tended to purchase food about as frequently as recipients of electronic transfers did.

D. Some recipients of physical transfers received small amounts of money relatively frequently, while others received large amounts relatively infrequently.

ID: 8545ccfe

Icebergs generally appear to be mostly white or blue, depending on how the ice reflects sunlight. Ice with air bubbles trapped in it looks white because much of the light reflects off the bubbles. Ice without air bubbles usually looks blue because the light travels deep into the ice and only a little of it is reflected. However, some icebergs in the sea around Antarctica appear to be green. One team of scientists hypothesized that this phenomenon is the result of yellow-tinted dissolved organic carbon in Antarctic waters mixing with blue ice to produce the color green.

Which finding, if true, would most directly weaken the team's hypothesis?

A. White ice doesn't change color when mixed with dissolved organic carbon due to the air bubbles in the ice.

B. Dissolved organic carbon has a stronger yellow color in Antarctic waters than it does in other places.

C. Blue icebergs and green icebergs are rarely found near each other.

D. Blue icebergs and green icebergs contain similarly small traces of dissolved organic carbon.

ID: dc87adf4

Barchester Towers is an 1857 novel by Anthony Trollope. In the novel, Trollope's portrayal of Dr. Proudie underscores the character's exaggerated sense of his own abilities: _____

Which quotation from *Barchester Towers* most effectively illustrates the claim?

A. "It must not…be taken as proved that Dr. Proudie was a man of great mental powers, or even of much capacity for business, for such qualities had not been required in him."

B. "[Dr. Proudie] was comparatively young, and had, as he fondly flattered himself, been selected as possessing such gifts, natural and acquired, as must be sure to recommend him to a yet higher notice."

C. "[Dr. Proudie's] residence in the metropolis, rendered necessary by duties thus entrusted to him, his high connexions, and the peculiar talents and nature of the man, recommended him to persons in power."

D. "[Dr. Proudie] was certainly possessed of sufficient tact to answer the purpose for which he was required without making himself troublesome."

ID: 3bfcb73b

An archaeological team led by Piotr Bieliński and Sultan al-Bakri found remnants of a 4,000-year-old Bronze Age board game at a site in Oman. Little is left of the game except a stone board, which is carved with a grid and has places to hold game pieces. Some scholars claim that the game was largely played by traders.

Which finding, if true, would most directly support the scholars' claim?

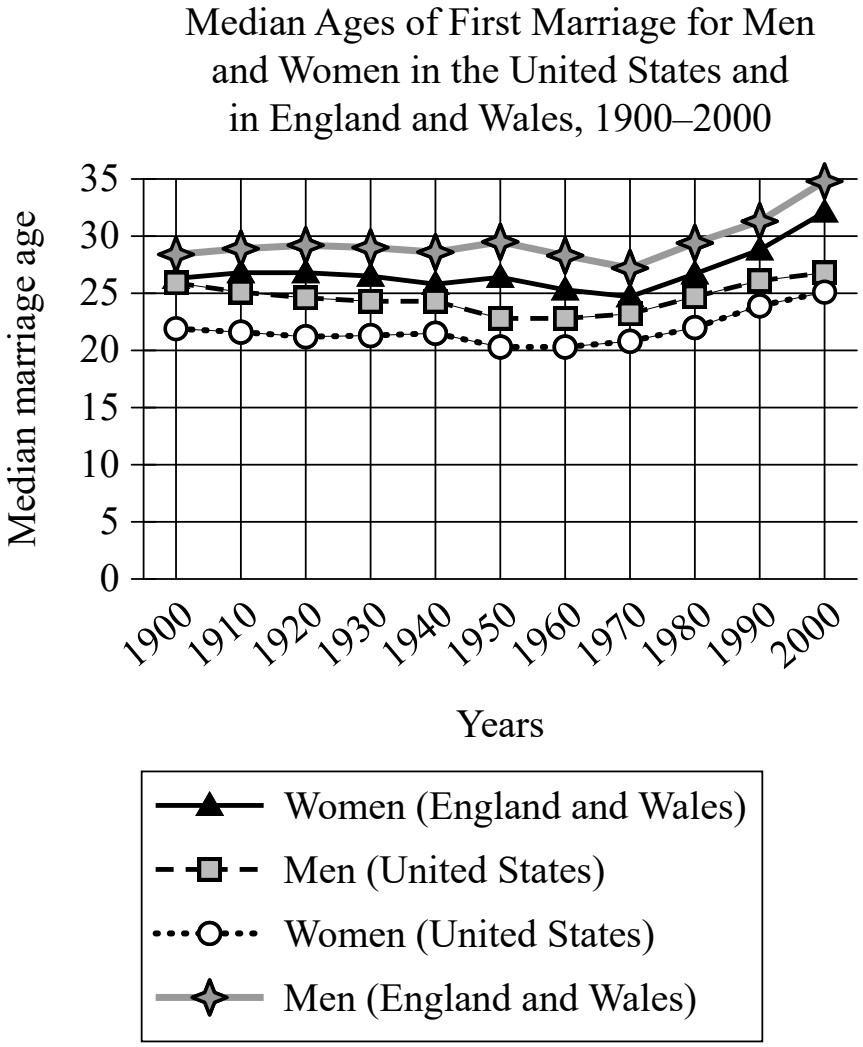
A. Other examples of the game dating to the same period have been found in the remains of several homes in the region, including in one home that may have belonged to a trader.

B. Similar games have been found in other sites dating to the same period that were connected to the site in Oman via trade routes.

C. The other known examples of the game dating to the same period have been found along routes that seem to have been used primarily by traders at the time.

D. Remnants of other goods have been found at the site in Oman that probably also reached the location through trade.

ID: 53c6c179



A sociology student is reading an essay on the median age of first marriage in Western countries throughout the twentieth century. The author of the essay cites factors common to these countries that the author believes caused an increase in the median age of first marriage, such as new technologies that shortened the time needed for domestic chores, making two-person households less necessary and living alone more viable. The student asserts that beyond these factors there must be additional ones specific to particular Western countries that influenced the increase of age at first marriage.

Which choice most effectively uses data from the graph that support the student's assertion?

A. Between 1970 and 2000, the median age of first marriage rose more sharply for men in England and Wales than it did for men in the United States.

B. In England and Wales, the median age of first marriage was consistently higher for men than for women between 1900 and 2000, but this was not always the case in the United States.

C. The median age of first marriage for men in England and Wales was lower in 1970 than in 1950 or 1990.

D. Between 1900 and 2000, the median age of first marriage for women in England and Wales was consistently higher than for women in the United States, as was the case for men.

ID: a44bbd6b

Several studies of sediment (e.g., dirt, pieces of rock, etc.) in streams have shown an inverse correlation between sediment grain size and downstream distance from the primary sediment source, suggesting that stream length has a sorting effect on sediment. In a study of sediment sampled at more than a dozen sites in Alpine streams, however, geologists Camille Litty and Fritz Schlunegger found that cross-site variations in grain size were not associated with differences in downstream distance, though they did not conclude that downstream distance is irrelevant to grain size. Rather, they concluded that sediment influx in these streams may have been sufficiently spatially diffuse to prevent the typical sorting effect from being observed.

Which finding about the streams in the study, if true, would most directly support Litty and Schlunegger's conclusion?

A. The streams regularly experience portions of their banks collapsing into the water at multiple points upstream of the sampling sites.

B. The streams contain several types of sediment that are not typically found in streams where the sorting effect has been demonstrated.

C. The streams mostly originate from the same source, but their lengths vary considerably due to the different courses they take.

D. The streams are fed by multiple tributaries that carry significant volumes of sediment and that enter the streams downstream of the sampling sites.

ID: 80fd9454

Percentage of Available Eggs Eaten by Cane Toad Tadpoles

Amphibian species (common name)	Percentage of eggs eaten	Native to Australia	Produces bufadienolide
Little red tree frog	1%	yes	no
Cane toad	90%	no	yes
Short-footed frog	7%	yes	no
Striped burrowing frog	10%	yes	no
Dainty green tree frog	1%	yes	no

Native to Latin America, the cane toad was introduced to Australia in the 1930s. In recent decades, tadpoles in the Australian population have been shown to consume eggs of their own species. A 2022 study showed that when presented with cane toad eggs as well as eggs of native Australian amphibians, cane toad tadpoles disproportionately consumed eggs of their own species. This behavior results from their attraction to bufadienolide, a chemical produced by the eggs of cane toads but not by the eggs of native amphibians. However, using data from this study, a student wishes to argue that the presence of bufadienolide doesn't entirely explain the cane toad tadpoles' preference for certain eggs over others.

Which choice best describes data from the table that support the student's argument?

A. The tadpoles consumed a higher percentage of the striped burrowing frog eggs than they did of the eggs of the dainty green tree frog.

B. The tadpoles left a certain percentage of the eggs of each of the five species unharmed, thus ultimately allowing them to hatch.

C. The tadpoles consumed a lower percentage of the short-footed frog eggs than they did of the eggs of their own species.

D. The tadpoles consumed the same percentage of the dainty green tree frog eggs as they did of the little red tree frog eggs.

ID: 63e7799d

In vertical inheritance, parents pass genes to their offspring, but in horizontal transfer (HT), one species, often bacteria, passes genetic material to an unrelated species. In a 2022 study, herpetologist Atsushi Kurabayashi and his team investigated HT in multicellular organisms—namely, snakes and frogs in Madagascar. The team detected *BovB*—a gene transmitted vertically in snakes—in many frog species. The apparent direction of gene transfer seems counterintuitive because frogs usually don't survive encounters with snakes and so wouldn't be able to transmit the newly acquired gene to offspring, but the team concluded that *BovB* is indeed transmitted from snakes to frogs, either directly or indirectly, via HT.

Which finding, if true, would most directly support the team's conclusion?

A. *BovB* can be transmitted across frog species through HT.

B. Parasites known to feed on species of snakes and frogs in which the *BovB* gene occurs also carry *BovB*.

C. *BovB* cannot be reliably transmitted from a snake species to bacteria that are usually encountered by frog species.

D. Frog species with *BovB* show few discernible advantages as compared with frog species that do not carry *BovB*.

ID: 860803dd

Biologist Valentina Gómez-Bahamón and her team have investigated two subspecies of the fork-tailed flycatcher bird that live in the same region in Colombia, but one subspecies migrates south for part of the year, and the other doesn't. The researchers found that, due to slight differences in feather shape, the feathers of migratory forked-tailed flycatcher males make a sound during flight that is higher pitched than that made by the feathers of nonmigratory males. The researchers hypothesize that fork-tailed flycatcher females are attracted to the specific sound made by the males of their own subspecies, and that over time the females' preference will drive further genetic and anatomical divergence between the subspecies.

Which finding, if true, would most directly support Gómez-Bahamón and her team's hypothesis?

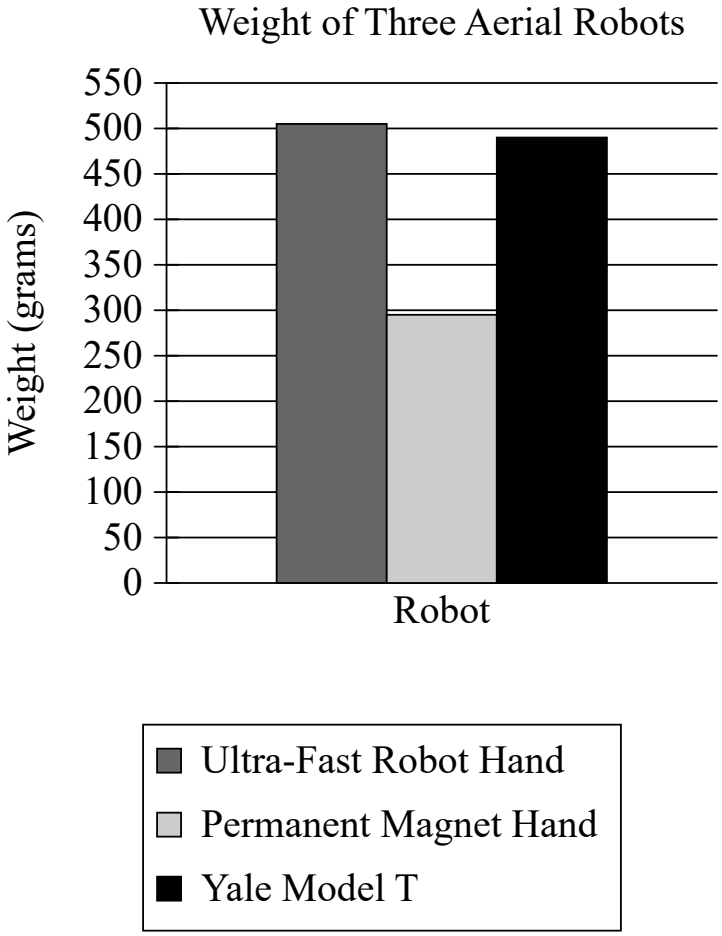
A. The feathers located on the wings of the migratory fork-tailed flycatchers have a narrower shape than those of the nonmigratory birds, which allows them to fly long distances.

B. Over several generations, the sound made by the feathers of migratory male fork-tailed flycatchers grows progressively higher pitched relative to that made by the feathers of nonmigratory males.

C. Fork-tailed flycatchers communicate different messages to each other depending on whether their feathers create high-pitched or low-pitched sounds.

D. The breeding habits of the migratory and nonmigratory fork-tailed flycatchers remained generally the same over several generations.

ID: df37c087



Aerial robots vary considerably in their holding force; the Ultra-Fast Robot Hand, for example, has a holding force of 56 newtons, more than twice that of the Permanent Magnet Hand and more than four times that of the Yale Model T. Since an aerial robot must lift its own weight along with its cargo, engineer Jiawei Meng and colleagues used a ratio of each robot's holding force to the robot's weight to calculate payload capacity, with higher ratios corresponding to greater capacity, concluding that the Ultra-Fast Robot Hand has a higher payload capacity than the Yale Model T.

Which choice best describes data in the graph that support Meng and colleagues' conclusion?

A. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than 450 grams.

B. The Ultra-Fast Robot Hand and the Yale Model T each weigh more than the Permanent Magnet Hand does.

C. The Yale Model T has a lower holding force than the Permanent Magnet Hand despite weighing more.

D. The Ultra-Fast Robot Hand weighs only slightly more than the Yale Model T does.

ID: f9c4bdab

A student is writing a paper about *One Night in Miami...*, a 2020 film directed by Regina King and written by Kemp Powers. Powers adapted the film's screenplay from his 2013 play, which he wrote after learning about a 1964 meeting that took place in Miami, Florida, between four prominent figures of the Civil Rights movement: Malcolm X, Muhammad Ali, Jim Brown, and Sam Cooke. The student claims that although Powers was inspired by this meeting, the film is best understood not as a precise retelling of historical events but rather as a largely imagined but informed representation of them.

Which quotation from an article about *One Night in Miami...* would be the most effective evidence for the student to include in support of this claim?

A. "When Powers learned of the meeting, he initially planned to write a much longer work about its four famous participants rather than focusing on the meeting itself."

B. "*One Night in Miami...* received numerous awards and nominations, including an Academy Award nomination for Powers for Best Adapted Screenplay."

C. "Powers has described *One Night in Miami...* as the story of four friends encouraging and supporting one another while engaged in a crucial political debate about how best to achieve equality for Black people in the United States."

D. "Powers could find only the most superficial historical details about the meeting, so he read extensively about the four individuals and their thinking at the time in an effort to portray what might have happened between them."

ID: b32c4b3a

The Intertropical Convergence Zone (ITCZ), a band of clouds that encircles Earth in the tropics and is a major rainfall source, shifts position in response to temperature variations across Earth's hemispheres. Data from Huagapo Cave in Peru suggest the ITCZ shifted south during the Little Ice Age (circa 1300–1850), but a shift as far into South America as Huagapo should have led to dry conditions in Central America, which is inconsistent with climate models. To resolve the issue, geologist Yemane Asmerom and colleagues collected data from Yok Balum Cave in Central America and compared them with the Huagapo data. They concluded that during the Little Ice Age, the ITCZ may have expanded northward and southward rather than simply shifted.

Which finding from Asmerom and colleagues' study, if true, would most directly support their conclusion?

A. Neither the Yok Balum data nor the Huagapo data show significant local variations in temperature during the Little Ice Age.

B. Both the Yok Balum data and the Huagapo data show increased temperatures and prolonged dry conditions during the Little Ice Age.

C. The Yok Balum data show prolonged dry conditions during the same portions of the Little Ice Age in which the Huagapo data show heightened levels of rainfall.

D. The Yok Balum data and the Huagapo data show strongly correlated patterns of high rainfall during the Little Ice Age.

ID: af9e3240

Electra is a circa 420–410 BCE play by Sophocles, translated in 1870 by R.C. Jebb. Electra, who is in mourning for her dead father and her long-absent brother, is aware of the intensity of her grief but believes it to be justified: _____

Which quotation from *Electra* most effectively illustrates the claim?

A. “O thou pure sunlight, and thou air, earth’s canopy, how often have ye heard the strains of my lament, the wild blows dealt against this bleeding breast, when dark night fails!”

B. “Send to me my brother; for I have no more the strength to bear up alone against the load of grief that weighs me down.”

C. “I know my own passion, it escapes me not; but, seeing that the causes are so dire, will never curb these frenzied plaints, while life is in me.”

D. “But never will I cease from dirge and sore lament, while I look on the trembling rays of the bright stars, or on this light of day.”

ID: 2c06139b

Tadpole Body Mass and Toxin Production after Three Weeks in Ponds

Population density	Average tadpole body mass (milligrams)	Average number of distinct bufadienolide toxins per tadpole	Average amount of bufadienolide per tadpole (nanograms)	Average bufadienolide concentration (nanograms per milligram of tadpole body mass)
High	193.87	22.69	5,815.51	374.22
Medium	254.56	21.65	5,525.72	230.10
Low	258.97	22.08	4,664.99	171.43

Ecologist Veronika Bókony and colleagues investigated within-species competition among common toads (*Bufo bufo*), a species that secretes various unpleasant-tasting toxins called bufadienolides in response to threats. The researchers tested *B. bufo* tadpoles' responses to different levels of competition by creating ponds with different tadpole population densities but a fixed amount of food. Based on analysis of the tadpoles after three weeks, the researchers concluded that increased competition drove bufadienolide production at the expense of growth.

Which choice uses data from the table to most effectively support the researchers' conclusion?

A. The difference in average tadpole body mass was small between the low and medium population density conditions and substantially larger between the low and high population density conditions.

B. Tadpoles in the low and medium population density conditions had substantially lower average bufadienolide concentrations but had greater average body masses than those in the high population density condition.

C. Tadpoles in the high population density condition displayed a relatively modest increase in the average amount of bufadienolide but roughly double the average bufadienolide concentration compared to those in the low population density condition.

D. Tadpoles produced approximately the same number of different bufadienolide toxins per individual across the population density conditions, but average tadpole body mass decreased as population density increased.

ID: 01c1d9ee

Swahili Speakers in Three African Countries

Country	Approximate number of speakers (in millions)	Estimated % of population
Democratic Republic of the Congo	22	25
Kenya	55	100
Tanzania	61	100

Swahili is estimated to be the first language of up to 15 million people worldwide. It's also an officially recognized language in Tanzania, Kenya, and the Democratic Republic of the Congo, which means these countries use Swahili in government documents and proceedings. But even in countries where almost everyone speaks Swahili, for many it isn't their first language but is instead their second, third, or even fourth language.

Which choice most effectively uses data from the table to support the underlined claim?

A. Tanzania has approximately 61 million Swahili speakers, which is much more than the estimated total number of people worldwide for whom Swahili is their first language.

B. Tanzania is estimated to have at most 15 million Swahili speakers, while the country's total population is approximately 61 million people.

C. Approximately 100 percent of the people who speak Swahili as their first language live in Kenya, which has a total population of approximately 55 million people.

D. Approximately 100 percent of Kenya's population speaks Swahili, while only about 25 percent of the Democratic Republic of the Congo's population speaks Swahili.

ID: b2e54b50

Correlations Between Congestion Ratings and Features of the Crowd in Raters' Immediate Vicinity

Crowd feature	Before obstacle	After obstacle	Overall
Density	0.8592	0.7308	0.7447
Velocity	−0.9357	−0.9518	−0.8587

Researcher Xiaolu Jia and colleagues monitored individuals' velocity and the surrounding crowd density as a group of study participants walked through a space and navigated around an obstacle. Participants rated how congested it seemed before the obstacle, after the obstacle, and overall, and the researchers correlated those ratings with velocity and density. (Correlations range from −1 to 1, with greater distance from 0 indicating greater strength). The researchers concluded that the correlations with velocity are stronger than those with density.

Which choice best describes data from the table that support the researchers' conclusion?

A. The correlation between congestion ratings before the obstacle and density is further from 0 than the correlation between overall congestion rating and velocity is.

B. The correlation between congestion ratings before the obstacle and velocity is further from 0 than the correlation between congestion overall and velocity is.

C. For each of the three ratings, the correlation with velocity is negative while the correlation with density is positive.

D. For each of the three ratings, correlations with velocity are further from 0 than the corresponding correlations with density are.

ID: 9c407117

A student performs an experiment testing her hypothesis that a slightly acidic soil environment is more beneficial for the growth of the plant *Brassica rapa parachinensis* (a vegetable commonly known as choy sum) than a neutral soil environment. She plants sixteen seeds of choy sum in a mixture of equal amounts of coffee grounds (which are highly acidic) and potting soil and another sixteen seeds in potting soil without coffee grounds as the control for the experiment. The two groups of seeds were exposed to the same growing conditions and monitored for three weeks.

Which finding, if true, would most directly weaken the student's hypothesis?

A. The choy sum planted in the soil without coffee grounds were significantly taller at the end of the experiment than the choy sum planted in the mixture of soil and coffee grounds.

B. The choy sum grown in the soil without coffee grounds weighed significantly less at the end of the experiment than the choy sum grown in the mixture of soil and coffee grounds.

C. The choy sum seeds planted in the soil without coffee grounds sprouted significantly later in the experiment than did the seeds planted in the mixture of soil and coffee grounds.

D. Significantly fewer of the choy sum seeds planted in the soil without coffee grounds sprouted plants than did the seeds planted in the mixture of soil and coffee grounds.

ID: 378c66d5

A member of the Otomi, an Indigenous people in Central Mexico, Octavio Medellín immigrated to the United States as a child, and his sculpture bears the impress of traditions on both sides of the border: US-based modernist sculpture, Mexican modernist painting, Otomi art, and the ancient sculpture of other Mexican Indigenous peoples, including the Maya. In his 1950 masterpiece *History of Mexico*, Medellín fuses these influences into a style so idiosyncratic that it resists efforts to view his work through the lens of nationality or cultural identity. Artists, he insisted, should strive for individual expression, even as they draw inspiration from their heritage and the communities where they live and work.

Which quotation from an art critic most directly challenges the underlined claim in the text?

A. "Although a number of ancient Indigenous artistic traditions pictured human forms in profile, the forms populating the surface of *A History of Mexico* suggest a specifically Maya influence."

B. "In *A History of Mexico*, the synthesis of ancient and modernist traditions functions as a stylistic parallel to the work's subject matter: a survey of centuries of Mexican history."

C. "Many critics focus on Indigenous influences in *A History of Mexico* and other key works by Medellín to the exclusion of influences from non-Indigenous art."

D. "While *A History of Mexico* features modernist motifs, it relies primarily on angular human forms in profile—a staple of Maya sculpture—and thus invites classification as Indigenous art."

ID: 35ec767c

Corn-Related Vocabulary in Various Southeastern Languages

Language family	Word (language)	English translation	Proposed origin in vocabulary of the Totozoquean language family
Muskogean	tanchi’ (Chickasaw); tanchi (Choctaw); vce (Muscogee, pronounced “uh-chi”)	corn	no
Iroquoian	se-lu (Cherokee)	corn	no
Caddoan	-k’as- (Caddo)	dried corn	yes
Chitimacha	k’asma (Chitimacha)	corn	yes

In Caddo, a language from what is now the US Southeast, vocabulary pertaining to corn cultivation resembles equivalent vocabulary in the Totozoquean language family in Mexico. This resemblance is perhaps attributable to cultural contact: such words could have entered Caddo through the intermediary of the neighboring but unrelated Chitimacha language, concurrent with the dissemination of corn itself from Mexico into the Southeast after 700 CE. That the vocabulary pertaining to domestic crops accompanies them as they diffuse into new regions is an established phenomenon globally. Crops may also be decoupled from vocabulary altogether: corn cultivation became ubiquitous among the Southeastern tribes, yet _____

Which choice most effectively uses data from the table to complete the statement?

A. the origins of vocabulary pertaining to the crop vary across languages in the region, with the words for corn in Cherokee and the Muskogean languages showing no demonstrable relationship to Totozoquean vocabulary.

B. the region is linguistically diverse, being home not only to Chitimacha and Caddo, but also to the Muskogean language family (including Chickasaw, Choctaw, and Muscogee) and to one Iroquoian language (Cherokee).

C. corn-related vocabulary underwent changes when entering other, unrelated languages, as can be seen by the divergence of the Caddo word from the Chitimacha word it originated in.

D. words for corn in the languages of the Muskogean family evolved from a common root, with the Muscogee word having lost certain consonant sounds still present in the Chickasaw and Choctaw words.

ID: 24c1b7e4

Percentage Point Changes in US Federal Outlays Relative to GDP by Congressional Status

Period	Congressional status	Change in total outlays	Change in nondefense outlays	Change in defense outlays
1981–1988	divided	−0.4	−1.3	0.9
1975–1976	divided	2.7	3.0	−0.3
1977–1980	undivided	0.3	0.6	−0.3
1964–1968	undivided	1.9	1.4	0.5
1969–1974	divided	−1.8	2.1	−3.9

Economist Steve H. Hanke has shown that divided US Congresses—which occur when one party holds the majority in the House of Representatives and another holds the majority in the Senate—tend to accompany reductions in total federal outlays (spending) relative to gross domestic product (GDP), which Hanke interprets to reflect decreases in government size. Hanke calculated the percentage point change in total outlays (encompassing nondefense and defense outlays) for consecutive US Congresses. Hanke has pointed to his calculations as evidence that a divided Congress may be a "necessary but not sufficient condition" for a decrease in government size to occur.

Which choice best describes data from the table that support the underlined claim?

A. The periods of undivided Congresses were associated with increases in nondefense outlays, whereas all the periods of divided Congresses except one were associated with reductions in defense outlays.

B. All the periods of divided Congresses were associated with reductions in total outlays, although two periods were also associated with increases in nondefense outlays.

C. The periods of undivided Congresses were associated with increases in total outlays, whereas all the periods of divided Congresses were associated with reductions in either nondefense outlays or defense outlays.

D. All the periods of divided Congresses except one were associated with reductions in total outlays, whereas the periods of undivided Congresses were associated with increases in total outlays.

ID: 56f477fb

Distribution of Ecosystem Services Affected by Invasive Species by Service Type

Region (Overall)	Provisioning (75%)	Regulating (21%)	Cultural (4%)
West	73%	27%	0%
North	88%	12%	0%
South	79%	14%	7%
East	83%	6%	11%
Central	33%	67%	0%

To assess the impact of invasive species on ecosystems in Africa, Benis N. Egoh and colleagues reviewed government reports from those nations about how invasive species are undermining ecosystem services (aspects of the ecosystem on which residents depend). The services were sorted into three categories: provisioning (material resources from the ecosystem), regulating (natural processes such as cleaning the air or water), and cultural (nonmaterial benefits of ecosystems). Egoh and her team assert that countries in each region reported effects on provisioning services and that provisioning services represent the majority of the reported services.

Which choice best describes data from the table that support Egoh and colleagues' assertion?

A. Provisioning services represent 73% of the services reported for the West region and 33% of those for the Central region, but they represent 75% of the services reported overall.

B. None of the percentages shown for provisioning services are lower than 33%, and the overall percentage shown for provisioning services is 75%.

C. Provisioning services are shown for each region, while no cultural services are shown for some regions.

D. The greatest percentage shown for provisioning services is 88% for the North region, and the least shown for provisioning services is 33% for the Central region.

ID: 4fc9a13a

The novelist Toni Morrison was the first Black woman to work as an editor at the publishing company Random House, from 1967 to 1983. A scholar asserts that one of Morrison's likely aims during her time as an editor was to strengthen the presence of Black writers on the list of Random House's published authors.

Which finding, if true, would most strongly support the scholar's claim?

A. The percentage of authors published by Random House who were Black rose in the early 1970s and stabilized throughout the decade.

B. Black authors who were interviewed in the 1980s and 1990s were highly likely to cite Toni Morrison's novels as a principal influence on their work.

C. The novels written by Toni Morrison that were published after 1983 sold significantly more copies and received wider critical acclaim than the novels she wrote that were published before 1983.

D. Works that were edited by Toni Morrison during her time at Random House displayed stylistic characteristics that distinguished them from works that were not edited by Morrison.

ID: 124fdcd7

Many archaeologists will tell you that categorizing excavated fragments of pottery by style, period, and what objects they belong to relies not only on standard criteria, but also on instinct developed over years of practice. In a recent study, however, researchers trained a deep-learning computer model on thousands of images of pottery fragments and found that it could categorize them as accurately as a team of expert archaeologists. Some archaeologists have expressed concern that they might be replaced by such computer models, but the researchers claim that outcome is highly unlikely.

Which finding, if true, would most directly support the researchers' claim?

A. In the researchers' study, the model was able to categorize the pottery fragments much more quickly than the archaeologists could.

B. In the researchers' study, neither the model nor the archeologists were able to accurately categorize all the pottery fragments that were presented.

C. A survey of archaeologists showed that categorizing pottery fragments limits the amount of time they can dedicate to other important tasks that only human experts can do.

D. A survey of archaeologists showed that few of them received dedicated training in how to properly categorize pottery fragments.

ID: f1bfbed3

Marta Coll and colleagues' 2010 Mediterranean Sea biodiversity census reported approximately 17,000 species, nearly double the number reported in Carlo Bianchi and Carla Morri's 2000 census—a difference only partly attributable to the description of new invertebrate species in the interim. Another factor is that the morphological variability of microorganisms is poorly understood compared to that of vertebrates, invertebrates, plants, and algae, creating uncertainty about how to evaluate microorganisms as species. Researchers' decisions on such matters therefore can be highly consequential. Indeed, the two censuses reported similar counts of vertebrate, plant, and algal species, suggesting that _____

Which choice most logically completes the text?

A. Coll and colleagues reported a much higher number of species than Bianchi and Morri did largely due to the inclusion of invertebrate species that had not been described at the time of Bianchi and Morri's census.

B. some differences observed in microorganisms may have been treated as variations within species by Bianchi and Morri but treated as indicative of distinct species by Coll and colleagues.

C. Bianchi and Morri may have been less sensitive to the degree of morphological variation displayed within a typical species of microorganism than Coll and colleagues were.

D. the absence of clarity regarding how to differentiate among species of microorganisms may have resulted in Coll and colleagues underestimating the number of microorganism species.

ID: ce4448b7

Researchers recently found that disruptions to an enjoyable experience, like a short series of advertisements during a television show, often increase viewers' reported enjoyment. Suspecting that disruptions to an unpleasant experience would have the opposite effect, the researchers had participants listen to construction noise for 30 minutes and anticipated that those whose listening experience was frequently interrupted with short breaks of silence would thus _____

Which choice most logically completes the text?

A. find the disruptions more irritating as time went on.

B. rate the listening experience as more negative than those whose listening experience was uninterrupted.

C. rate the experience of listening to construction noise as lasting for less time than it actually lasted.

D. perceive the volume of the construction noise as growing softer over time.

ID: 58e9e497

In the early nineteenth century, some Euro-American farmers in the northeastern United States used agricultural techniques developed by the Haudenosaunee (Iroquois) people centuries earlier, but it seems that few of those farmers had actually seen Haudenosaunee farms firsthand. Barring the possibility of several farmers of the same era independently developing techniques that the Haudenosaunee people had already invented, these facts most strongly suggest that _____

Which choice most logically completes the text?

A. those farmers learned the techniques from other people who were more directly influenced by Haudenosaunee practices.

B. the crops typically cultivated by Euro-American farmers in the northeastern United States were not well suited to Haudenosaunee farming techniques.

C. Haudenosaunee farming techniques were widely used in regions outside the northeastern United States.

D. Euro-American farmers only began to recognize the benefits of Haudenosaunee farming techniques late in the nineteenth century.

ID: f9bd4e61

German theater practitioner Bertolt Brecht (1898–1956) believed that theater should elicit an intellectual rather than an emotional response from audiences, provoking them to consider social and political realities that extend beyond the characters and events depicted onstage. Brecht's influence can be seen in English playwright Caryl Churchill's 1979 play *Cloud 9*: although the play sometimes invites empathetic reactions, it primarily works to engage audiences in an interrogation of patriarchy and colonialism, which it does by placing audiences at a distance, thereby encouraging them to _____

Which choice most logically completes the text?

A. focus on the characters' beliefs about social and political issues as revealed by the characters' actions.

B. reflect on social and political phenomena not directly related to patriarchy and colonialism.

C. recognize pertinent social and political parallels between Germany during Brecht's time and England at the time when Churchill was writing *Cloud 9*.

D. be dispassionate as they think critically about the social and political questions raised by the play.

ID: 4889580c

Archaeologists and historians used to believe that the Maya civilization during its Classic period (roughly 250–900) lacked agricultural marketplaces. One reason for this belief was that these scholars misunderstood the ecology of the regions the Maya inhabited. Marketplaces typically emerge because different individuals or groups want to trade resources they control for resources they don't control. Scholars seriously underestimated the ecological diversity of the Maya landscape and thus assumed that _____

Which choice most logically completes the text?

A. marketplaces likely would not have attracted many traders from outside the regions controlled by the Maya.

B. farming practices would have been largely the same throughout Maya lands even if the crops people produced varied significantly.

C. marketplaces would not have enabled Maya people to acquire many products different from those they already produced.

D. farmers would trade agricultural products only if they had already produced enough to meet their own needs.

ID: aaddd60f

Scientists studying Mars long thought the history of its crust was relatively simple. One reason for this is that geologic and climate data collected by a spacecraft showed that the crust was largely composed of basalt, likely as a result of intense volcanic activity that brought about a magma ocean, which then cooled to form the planet's surface. A study led by Valerie Payré focused on additional information—further analysis of data collected by the spacecraft and infrared wavelengths detected from Mars's surface—that revealed the presence of surprisingly high concentrations of silica in certain regions on Mars. Since a planetary surface that formed in a mostly basaltic environment would be unlikely to contain large amounts of silica, Payré concluded that _____

Which choice most logically completes the text?

A. the information about silica concentrations collected by the spacecraft is likely more reliable than the silica information gleaned from infrared wavelengths detected from Mars's surface.

B. high silica concentrations on Mars likely formed from a different process than that which formed the crusts of other planets.

C. having a clearer understanding of the composition of Mars's crust and the processes by which it formed will provide more insight into how Earth's crust formed.

D. Mars's crust likely formed as a result of other major geological events in addition to the cooling of a magma ocean.

ID: 08395130

The Hubble Space Telescope (HST) is projected to maintain operation until at least 2030, but it has already revolutionized high-resolution imaging of solar-system bodies in visible and ultraviolet (UV) light wavelengths, notwithstanding that only about 6% of the bodies imaged by the HST are within the solar system. NASA researcher Cindy L. Young and colleagues assert that a new space telescope dedicated exclusively to solar-system observations would permit an extensive survey of minor solar-system bodies and long-term UV observation to discern how solar-system bodies change over time. Young and colleagues' recommendation therefore implies that the HST _____

Which choice most logically completes the text?

A. will likely continue to be used primarily to observe objects outside the solar system.

B. will no longer be used to observe solar system objects if the telescope recommended by Young and colleagues is deployed.

C. can be modified to observe the features of solar system objects that are of interest to Young and colleagues.

D. lacks the sensors to observe the wavelengths of light needed to discern how solar system bodies change over time.

ID: cef77aa7

Geoglyphs are large-scale designs of lines or shapes created in a natural landscape. The Nazca Lines were created in the Nazca Desert in Peru by several Indigenous civilizations over a period of many centuries. Peruvian archaeologist Johny Isla specializes in these geoglyphs. At a German exhibit about the Nazca Lines, he saw an old photograph of a large geoglyph of a whalelike figure and was surprised that he didn't recognize it. Isla returned to Peru and used a drone to search a wide area, looking for the figure from the air. This approach suggests that Isla thought that if he hadn't already seen it, the whalelike geoglyph _____

Which choice most logically completes the text?

A. must represent a species of whale that went extinct before there were any people in Peru.

B. is actually located in Germany, not Peru, and isn't part of the Nazca Lines at all.

C. is probably in a location Isla hadn't ever come across while on the ground.

D. was almost certainly created a long time after the other Nazca Lines geoglyphs were created.

ID: 9c591ff7

Some *Astyanax mexicanus*, a river-dwelling fish found in northeast Mexico, have colonized caves in the region. Although there is little genetic difference between river and cave *A. mexicanus* and all members of the species can emit the same sounds, biologist Carole Hyacinthe and colleagues found that the context and significance of those sounds vary by location—e.g., the click that river-dwelling *A. mexicanus* use to signal aggression is used by cave dwellers when foraging—and the acoustic properties of cave fish sounds show some cave-specific variations as well. Hyacinthe and colleagues note that differences in sonic communication could accumulate to the point of inhibiting interbreeding among fish from different locations, suggesting that _____

Which choice most logically completes the text?

A. although *A. mexicanus* living in rivers are genetically similar to those living in caves, river fish rely on sonic communication less than cave fish do.

B. although *A. mexicanus* is a single species at present, it could be in the process of splitting into distinct populations with different characteristics.

C. although all *A. mexicanus* emit sounds, the fish living in rivers produce some sounds that the fish living in caves do not, and vice versa.

D. although *A. mexicanus* from different locations can interbreed currently, river fish and cave fish are sufficiently genetically distinct that they can be considered separate species.

ID: 2a075bd1

Indigenous cultures possess unique knowledge of the medicinal uses of plants. According to a 2021 study, 73 percent of the medicinal uses of plants native to North America are reflected in the vocabulary of a single Indigenous language. However, as more and more Indigenous people exclusively speak a globally dominant language, such as English, their ancestral languages fade from daily use. These facts lend added importance to tribal nations' efforts to preserve their languages. By ensuring the continued use of Cherokee, Ojibwe, and the hundreds of other Indigenous languages in what is now the United States, tribal nations are also _____

Which choice most logically completes the text?

A. increasing the number of medicinal plants represented in the vocabularies of Indigenous languages.

B. transmitting terms for medicinal plants from Indigenous languages to globally dominant languages.

C. preserving knowledge about the medicinal value of plants native to the tribal nations' lands.

D. ensuring that citizens of tribal nations have physical access to medicinal plants.

ID: 95dbdf51

Laura Mulvey has theorized that in narrative film, shots issuing from a protagonist's point of view compel viewers to identify with the character. Such identification is heightened by "invisible editing," or editing so inconspicuous that it renders cuts between shots almost unnoticeable. Conversely, Mulvey proposes that conspicuous editing or an absence of point-of-view shots would induce a more critical stance toward a protagonist. Consider, for example, the attic scene in Alfred Hitchcock's *The Birds*, a conspicuously edited sequence of tens of shots, few of which correspond to the protagonist's point of view. According to Mulvey's logic, this scene should affect viewers by _____

Which choice most logically completes the text?

A. obscuring their awareness of the high degree of artifice involved in constructing the montage.

B. lessening their identification with the protagonist, if not alienating them from the character altogether.

C. compelling them to identify with the film's director, whose proxy is the camera, and not with the protagonist.

D. diverting their attention away from the film's content and toward its stylistic attributes.

ID: a13c1c66

Many animals, including humans, must sleep, and sleep is known to have a role in everything from healing injuries to encoding information in long-term memory. But some scientists claim that, from an evolutionary standpoint, deep sleep for hours at a time leaves an animal so vulnerable that the known benefits of sleeping seem insufficient to explain why it became so widespread in the animal kingdom. These scientists therefore imply that _____

Which choice most logically completes the text?

A. prolonged deep sleep is likely advantageous in ways that have yet to be discovered.

B. most traits perform functions that are hard to understand from an evolutionary standpoint.

C. it is more important to understand how widespread prolonged deep sleep is than to understand its function.

D. many traits that provide significant benefits for an animal also likely pose risks to that animal.

ID: f942646f

Researchers Suchithra Rajendran and Maximilian Popfinger modeled varying levels of passenger redistribution from short-haul flights (flights of 50 to 210 minutes, from takeoff to landing) to high-speed rail trips. Planes travel faster than trains, but air travel typically requires 3 hours of lead time for security, baggage handling, and boarding that rail travel doesn't, so short-haul routes take similar amounts of time by air and by rail. However, the model suggests that as rail passenger volumes approach current capacity limits, long lead times emerge. Therefore, for rail to remain a viable alternative to short-haul flights, _____

Which choice most logically completes the text?

A. rail systems should offer fewer long-haul routes and airlines should offer more long-haul routes.

B. rail systems may need to schedule additional trains for these routes.

C. security, baggage handling, and boarding procedures used by airlines may need to be implemented for rail systems.

D. passengers who travel by rail for these routes will need to accept that lead times will be similar to those for air travel.

ID: 0dccbf17

Henry Ossawa Tanner's 1893 painting *The Banjo Lesson*, which depicts an elderly man teaching a boy to play the banjo, is regarded as a landmark in the history of works by Black artists in the United States. Scholars should be cautious when ascribing political or ideological values to the painting, however: beliefs and assumptions that are commonly held now may have been unfamiliar to Tanner and his contemporaries, and vice versa. Scholars who forget this fact when discussing *The Banjo Lesson* therefore _____

Which choice most logically completes the text?

A. risk judging Tanner's painting by standards that may not be historically appropriate.

B. tend to conflate Tanner's political views with those of his contemporaries.

C. forgo analyzing Tanner's painting in favor of analyzing his political activity.

D. wrongly assume that Tanner's painting was intended as a critique of his fellow artists.

ID: 61228830

A heliograph is a semaphore device used for sending optical communications—usually in the form of Morse code—by reflecting flashes of sunlight off a mirror. Heliographs were used for rapid communication across expansive distances for military, surveying, and forestry purposes during the late nineteenth and early twentieth centuries, but they were largely effective only during the daytime, and the range of the device depended on factors such as the opacity of the air and line of sight. Therefore, heliographs were eventually replaced by technology that _____

Which choice most logically completes the text?

A. worked on similar principles but was easier to produce and maintain.

B. was not so constrained by environmental circumstances.

C. could be used for more than military, surveying, or forestry purposes.

D. enabled communication that didn't require knowledge of Morse code.

ID: 6e0e0de1

Aerogels are highly porous foams consisting mainly of tiny air pockets within a solidified gel. These lightweight materials are often applied to spacecraft and other equipment required to withstand extreme conditions, as they provide excellent insulation despite typically being brittle and eventually fracturing due to degradation from repeated exposure to high heat. Now, Xiangfeng Duan of the University of California, Los Angeles, and colleagues have developed an aerogel with uniquely flexible properties. Unlike earlier aerogels, Duan's team's material contracts rather than expands when heated and fully recovers after compressing to just 5% of its original volume, suggesting that _____

Which choice most logically completes the text?

A. the aerogel's remarkable flexibility results from its higher proportion of air pockets to solidified gel as compared to other aerogels.

B. the aerogel's overall strength is greater than that of other insulators but its ability to withstand exposure to intense heat is lower.

C. the aerogel will be more effective as an insulator for uses that involve gradual temperature shifts than for those that involve rapid heat increases.

D. the aerogel will be less prone to the structural weakness that ultimately causes most other aerogels to break down with use.

ID: d1539546

Tides can deposit large quantities of dead vegetation within a salt marsh, smothering healthy plants and leaving a salt panne—a depression devoid of plants that tends to trap standing water—in the marsh's interior. Ecologist Kathryn Beheshti and colleagues found that burrowing crabs living within these pannes improve drainage by loosening the soil, leading the pannes to shrink as marsh plants move back in. At salt marsh edges, however, crab-induced soil loosening can promote marsh loss by accelerating erosion, suggesting that the burrowing action of crabs _____

Which choice most logically completes the text?

A. can be beneficial to marshes with small pannes but can be harmful to marshes with large pannes.

B. may promote increases in marsh plants or decreases in marsh plants, depending on the crabs' location.

C. tends to be more heavily concentrated in areas of marsh interiors with standing water than at marsh edges.

D. varies in intensity depending on the size of the panne relative to the size of the surrounding marsh.

ID: 9abc3ba5

“Gestures” in painting are typically thought of as bold, expressive brushstrokes. In the 1970s, American painter Jack Whitten built a 12-foot (3.7-meter) tool he named the “developer” to apply paint to an entire canvas in one motion, resulting in his series of “slab” paintings from that decade. Whitten described this process as making an entire painting in “one gesture,” signaling a clear departure from the prevalence of gestures in his work from the 1960s. Some art historians claim this shift represents “removing gesture” from the process. Therefore, regardless of whether using the developer constitutes a gesture, both Whitten and these art historians likely agree that _____

Which choice most logically completes the text?

A. any tool that a painter uses to create an artwork is capable of creating gestures.

B. Whitten’s work from the 1960s exhibits many more gestures than his work from the 1970s does.

C. Whitten became less interested in exploring the role of gesture in his work as his career progressed.

D. Whitten’s work from the 1960s is much more realistic than his work from the 1970s is.

ID: 22b3da87

During the Bourbon Restoration in France (1814–1830), the right to vote required in part that a person paid at least 300 francs in direct taxes to the government. The four most common taxes (the *quatre vieilles*) were levied on real estate (both land and buildings); the doors and windows in taxpayer homes; the rental values of homes; and the businesses of artisans and merchants. (Foreign investments were either exempt from taxation or taxed lightly.) Although relatively few people paid the tax on real estate, it was the main means of voter qualification and accounted for over two-thirds of government receipts during this period, suggesting that during the Bourbon Restoration _____

Which choice most logically completes the text?

A. those people who had the right to vote most likely had substantial holdings of French real estate.

B. the voting habits of French artisans and merchants were effective in reducing tax burdens on businesses.

C. the number of doors and windows in French residences was kept to a minimum but increased after 1830.

D. French people with significant foreign investments were unlikely to have the right to vote.

ID: cae97f58

Mosses can struggle in harsh desert conditions because these plants require enough sunlight for photosynthesis but not so much that they risk drying out. Researchers Jenna Ekwealor and Kirsten M. Fisher found several species of *Syntrichia caninervis*, a type of desert moss, growing under quartz crystals in California's Mojave Desert. To evaluate whether these semitransparent rocks benefited the moss, the researchers compared the shoot tissue, a measure of plant growth, of *S. caninervis* when growing on the soil surface versus when the moss was growing under the quartz rocks. They found that the shoot tissue was 62% longer for moss growing under the quartz as compared to moss on the soil surface, suggesting that _____

Which choice most logically completes the text?

A. *S. caninervis* is one of the few types of moss that can survive under semitransparent rocks.

B. quartz crystals do not transmit the necessary sunlight for photosynthesis in *S. caninervis*.

C. *S. caninervis* growing under quartz crystals experience lower light intensity and are thus able to retain more moisture.

D. quartz crystals are capable of supporting *S. caninervis* growth if the crystals are not too thin.

ID: 03701ef3

To better understand the burrowing habits of *Alpheus bellulus* (the tiger pistol shrimp), some studies have used resin casting to obtain precise measurements of the shrimps' burrows. Resin casting involves completely filling an empty burrow with a liquid plastic that hardens to create a three-dimensional model; however, recovering the model inevitably requires destroying the burrow. In their 2022 study, Miyu Umehara and colleagues discovered that an x-ray computed tomography (CT) scanner can accurately record a burrow's measurements both at a moment in time and throughout the entire burrow-building process, something that's impossible with resin casting because _____

Which choice most logically completes the text?

A. it can only be used on burrows below a certain size.

B. it does not allow for multiple castings of the same burrow over time.

C. the casting process takes more time than *A. bellulus* takes to construct a burrow.

D. the process of recovering the model distorts the resin's shape.

ID: ed314256

The most recent iteration of the immersive theater experience *Sleep No More*, which premiered in New York City in 2011, transforms its performance space—a five-story warehouse—into a 1930s-era hotel. Audience members, who wander through the labyrinthine venue at their own pace and follow the actors as they play out simultaneous, interweaving narrative loops, confront the impossibility of experiencing the production in its entirety. The play's refusal of narrative coherence thus hinges on the sense of spatial fragmentation that the venue's immense and intricate layout generates.

What does the text most strongly suggest about *Sleep No More*'s use of its performance space?

A. The choice of a New York City venue likely enabled the play's creators to experiment with the use of theatrical space in a way that venues from earlier productions could not.

B. Audience members likely find the experience of the play disappointing because they generally cannot make their way through the entire venue.

C. The production's dependence on a particular performance environment would likely make it difficult to reproduce exactly in a different theatrical space.

D. Audience members who navigate the space according to a recommended itinerary will likely have a better grasp of the play's narrative than audience members who depart from that itinerary.

ID: 1a2b29c9

The following text is adapted from María Cristina Mena's 1914 short story "The Vine-Leaf."

> It is a saying in the capital of Mexico that Dr. Malsufrido carries more family secrets under his hat than any archbishop. The doctor's hat is, appropriately enough, uncommonly capacious, rising very high, and sinking so low that it seems to be supported by his ears and eyebrows, and it has a furry look, as if it had been brushed the wrong way, which is perhaps what happens to it if it is ever brushed at all. When the doctor takes it off, the family secrets do not fly out like a flock of parrots, but remain nicely bottled up beneath a dome of old and highly polished ivory.

Based on the text, how do people in the capital of Mexico most likely regard Dr. Malsufrido?

A. Many have come to tolerate him despite his disheveled appearance.

B. Few feel concerned that he will divulge their confidences.

C. Some dislike how freely he discusses his own family.

D. Most would be unimpressed by him were it not for his professional expertise.

ID: 2fdfe002

The following text is adapted from Countee Cullen's 1926 poem "Thoughts in a Zoo."

> They in their cruel traps, and we in ours,
> Survey each other's rage, and pass the hours
> Commiserating each the other's woe,
> To mitigate his own pain's fiery glow.
> Man could but little proffer in exchange
> Save that his cages have a larger range.
> That lion with his lordly, untamed heart
> Has in some man his human counterpart,
> Some lofty soul in dreams and visions wrapped,
> But in the stifling flesh securely trapped.

Based on the text, what challenge do humans sometimes experience?

A. They cannot effectively tame certain wild animals because of a lack of compassion.

B. They cannot focus on setting attainable goals because of a lack of motivation.

C. They quickly become frustrated when faced with difficult tasks because of a lack of self-control.

D. They have aspirations that cannot be fulfilled because of certain limitations.

ID: 701126bc

In superfluorescence, electrical charges known as dipoles emit light in synchronized bursts so intense that they are visible to the eye. Until recently, this phenomenon has only been observed at extremely cold temperatures because dipoles cannot synchronize at higher temperatures. But in a study, Melike Biliroglu and colleagues observed superfluorescence at room temperature in thin films made of perovskite and other similarly crystalline materials; the researchers propose that the formation of shock-absorbing quasiparticles called polarons in the material protects dipoles from thermal interference.

Based on the text, how are polarons believed to be involved in the superfluorescence observed in Biliroglu and colleagues' study?

A. Polarons enable superfluorescent bursts to cross from one crystalline material to another.

B. Polarons allow for the dipoles to synchronize despite higher temperatures.

C. Polarons accelerate the dipoles' release of superfluorescent bursts.

D. Polarons decrease the intensity of the superfluorescent burst.

ID: 9731a22b

Paleontologist Lucas E. Fiorelli and colleagues have reported the discovery at a mine in Brazil of several egg clutches, partially preserved single eggs, and egg shells from the Late Cretaceous period. The researchers have concluded that the area was once a nesting and breeding site for titanosaurs, a group of sauropod dinosaurs. The finding is significant given the previous lack of known nesting sites in northern regions of South America, which led many paleontologists to assume that titanosaurs migrated south to lay eggs.

What does the text most strongly suggest about the site discovered by the researchers?

A. It is the earliest known example of a titanosaur nesting and breeding site.

B. It was very difficult to excavate given that it was discovered in a mine.

C. It may have been occupied by other sauropods in addition to titanosaurs.

D. It is farther north than any other nesting site discovered in South America.

ID: f7bd14de

Several scholars have argued that conditions in England in the late ninth through early eleventh centuries—namely, burgeoning literacy amid running conflicts between England's Anglo-Saxon kingdoms and Danish invaders—were especially conducive to the production of the Old English epic poem *Beowulf*, and they have dated the poem's composition accordingly. It is not inconceivable that *Beowulf* emerged from such a context, but privileging contextual fit over the linguistic evidence of an eighth- or even seventh-century composition requires a level of justification that thus far has not been presented.

Which choice best states the main idea of the text?

A. Although there are some grounds for believing that *Beowulf* was composed between the late ninth and early eleventh centuries, advocates for that view tend to rely on evidence that has been called into question by advocates for an earlier date.

B. Although several scholars have dated *Beowulf* to the late ninth through early eleventh centuries, others have argued that doing so privileges a controversial interpretation of the social conditions of the period.

C. Although *Beowulf* fits well with the historical context of England in the late ninth through early eleventh centuries, it fits equally well with the historical context of England in the seventh and eighth centuries.

D. Although the claim of a late ninth- through early eleventh-century composition date for *Beowulf* has some plausibility, advocates for the claim have not compellingly addressed evidence suggesting an earlier date.

ID: 96802cc0

For centuries, the widespread acknowledgment of the involvement of the cerebellum—a dense brain structure in vertebrates—in coordinating motor control in humans has hindered recognition of other possible functions of the structure. Neuroscience research from the last two decades now suggests that the cerebellum regulates emotion and social behavior, and recent research by Ilaria Carta and colleagues has identified a pathway connecting the cerebellum to a center for motivation and reward processing known as the ventral tegmental area (VTA).

Which choice best states the main idea of the text?

A. The recent verification of a pathway between the VTA and the cerebellum confirms the cerebellum's long-suspected role in motor coordination.

B. Recent advances in the field of neuroscience have challenged widely accepted claims about the function of a pathway connecting the VTA and the cerebellum.

C. The cerebellum has primarily been thought to regulate motor functioning, but in recent years neuroscience researchers have been uncovering additional functions.

D. Technological limitations have historically hindered the study of the cerebellum, but the recent development of new technologies has led to greater insights into its functions.

ID: d0f51067

Modern dog breeds are largely the result of 160 years of owners crossbreeding certain dogs in order to select for particular physical appearances. Owners often say that some breeds are also more likely than others to have particular personality traits—basset hounds are affectionate; boxers are easy to train—but Kathleen Morrill and colleagues found through a combination of owner surveys and DNA sequencing of 2,000 dogs that while physical traits are predictably heritable among purebred dogs, behavior varies widely among dogs of the same breed.

Which choice best states the main idea of the text?

A. Dog breeds would not exist without many years of human intervention in dogs' reproduction.

B. Research fails to confirm a commonly held belief about dog breeds and behavior.

C. The dog breeds most popular among owners have often changed over the past 160 years.

D. A study of dog breeds is notable for its usage of both opinion surveys and DNA sequencing.

ID: d0fbf1ae

Algae living within the tissues of corals play a critical role in keeping corals, and the marine ecosystems they are part of, thriving. Some coral species appear brown in color when healthy due to the algae colonies living in their tissues. In the event of an environmental stressor, the algae can die or be expelled, causing the corals to appear white. To recover the algae, the bleached corals then begin to produce bright colors, which block intense sunlight, encouraging the light-sensitive algae to recolonize the corals.

What does the text most strongly suggest about corals that produce bright colors?

A. These corals have likely been subjected to stressful environmental conditions.

B. These corals are likely more vulnerable to exposure from intense sunlight than white corals are.

C. These corals have likely recovered from an environmental event without the assistance of algae colonies.

D. These corals are more likely to survive without algae colonies than brown corals are.

ID: 14189fbb

Having written the impassioned call to arms “Letter to the Spanish Americans” in 1791, Peruvian intellectual Juan Pablo Viscardo y Guzmán is often considered a forerunner for the independence movements in Latin America. But Viscardo’s role in history would have remained insignificant were it not for Venezuelan revolutionary Francisco de Miranda, who was handed the unpublished letter after Viscardo’s death. Miranda not only helped circulate the letter, but his edits and footnotes to the text position Miranda as a central figure in the text’s creation.

Which choice best states the main idea of the text?

A. The original authorship of “Letter to the Spanish Americans” is disputed by contemporary historians.

B. The majority of the most eloquently stated arguments in “Letter to the Spanish Americans” were written by Miranda.

C. Miranda played a crucial role in influencing the content and distribution of “Letter to the Spanish Americans.”

D. “Letter to the Spanish Americans” persuaded many people in Latin America to pursue national independence.

ID: 4d3e3c52

In a paper about p-i-n planar perovskite solar cells (one of several perovskite cell architectures designed to collect and store solar power), Lyndsey McMillon-Brown et al. describe a method for fabricating the cell's electronic transport layer (ETL) using a spray coating. Conventional ETL fabrication is accomplished using a solution of nanoparticles. The process can result in a loss of up to 80% of the solution, increasing the cost of manufacturing at scale—an issue that may be obviated by spray coating fabrication, which the researchers describe as "highly reproducible, concise, and practical."

What does the text most strongly suggest about conventional ETL fabrication?

A. It is less suitable for manufacturing large volumes of planar p-i-n perovskite solar cells than an alternative fabrication method may be.

B. It is more expensive when manufacturing at scale than are processes for fabricating ETLs used in other perovskite solar cell architectures.

C. It typically entails a greater loss of nanoparticle solution than do other established approaches for ETL fabrication.

D. It is somewhat imprecise and therefore limits the potential effectiveness of p-i-n planar perovskite solar cells at capturing and storing solar power.

ID: d1b8a9ad

Disco remains one of the most ridiculed popular music genres of the late twentieth century. But as scholars have argued, the genre is far less superficial than many people believe. Take the case of disco icon Donna Summer: she may have been associated with popular songs about love and heartbreak (subjects hardly unique to disco, by the way), but like many Black women singers before her, much of her music also reflects concerns about community and identity. These concerns are present in many of the genre's greatest songs, and they generally don't require much digging to reveal.

What does the text most strongly suggest about the disco genre?

A. It has been unjustly ignored by most scholars despite the importance of the themes addressed by many of the genre's songs.

B. It evolved over time from a superficial genre focused on romance to a genre focused on more serious concerns.

C. It has been unfairly dismissed for the inclusion of subject matter that is also found in other musical genres.

D. It gave rise to a Black women's musical tradition that has endured even though the genre itself faded in the late twentieth century.

ID: db2da2bf

In 2019, 20 previously unknown moons were confirmed to be orbiting Saturn. Three of the moons have prograde orbits (orbiting in the direction the planet spins), and the other 17 have retrograde orbits (orbiting in the opposite direction of the planet's spin). All but one of the 20 moons are thought to be remnants of bodies that orbited Saturn until they broke apart in collisions. Although the one exceptional moon orbits in the same direction as the planet's spin, its orbit is highly eccentric compared to the rest, which may suggest that it has a different origin than the other 19 moons.

Based on the text, which choice best describes the moon with the eccentric orbit?

A. It doesn't have a retrograde orbit, but it likely has the same origin as the moons with retrograde orbits.

B. Its orbit is so tilted with respect to the other moons' orbits that it's neither prograde nor retrograde.

C. It has a prograde orbit that is likely the result of having collided with another body orbiting Saturn.

D. It has a prograde orbit and may not be a remnant of an earlier body that orbited Saturn.

ID: d2e0cba5

In a study of new technology adoption, Davit Marikyan et al. examined negative disconfirmation (which occurs when experiences fall short of one's expectations) to determine whether it could lead to positive outcomes for users. The team focused on established users of "smart home" technology, which presents inherent utilization challenges but tends to attract users with high expectations, often leading to feelings of dissonance. The researchers found that many users employed cognitive mechanisms to mitigate those feelings, ultimately reversing their initial sense of disappointment.

Which choice best states the main idea of the text?

A. Research suggests that most users of smart home technology will not achieve a feeling of satisfaction given the utilization challenges of such technology.

B. Although most smart home technology is aimed at meeting or exceeding users' high expectations, those expectations in general remain poorly understood.

C. Research suggests that users with high expectations for a new technology can feel content with that technology even after experiencing negative disconfirmation.

D. Although negative disconfirmation has often been studied, little is known about the cognitive mechanisms shaping users' reactions to it in the context of new technology adoption.

ID: 7c9a65bb

Optical tweezers are specialized scientific tools—particularly useful in biology and medicine—that use high-powered beams of light to trap and manipulate minuscule particles for study. Use of the tool has led to several scientific and medical breakthroughs over the last few decades, but the particles are often under prolonged exposure to the intense heat of the light beams. To overcome the risk of overheating, and thereby damage, researchers sometimes attach nano-sized glass beads to particles, allowing the light to focus on the beads instead of the particles.

Based on the text, what is one advantage of attaching glass beads to particles when using optical tweezers?

A. It decreases the time it takes for the optical tweezers to locate and capture the particles.

B. It facilitates the maneuvering of particles without directly heating the particles themselves.

C. It allows researchers to use weaker light beams to manipulate particles.

D. It adds a material to which particles can transfer any heat absorbed from the optical tweezers' light beam.

ID: a3fb5e77

Some animal-behavior studies involve observing wild animals in their natural habitat, and some involve capturing wild animals and observing them in a laboratory. Each approach has advantages over the other. In wild studies, researchers can more easily presume that the animals are behaving normally, and in lab studies, researchers can more easily control factors that might affect the results. But if, for example, the results from a wild study and a lab study of Western scrub-jays (*Aphelocoma californica*) contradict each other, one or both of the studies must have failed to account for some factor that was relevant to the birds' behavior.

Which choice best states the main idea of the text?

A. When the results of a natural-habitat study and those from a lab study of a wild animal such as the Western scrub-jay conflict, the study in the natural habitat is more likely than the lab study to have accurate results.

B. Studying wild animals such as the Western scrub-jay in both their natural habitat and lab settings is likely to yield conflicting results that researchers cannot fully resolve.

C. Wild animals such as the Western scrub-jay can be effectively studied in their natural habitat and in the lab, but each approach has drawbacks that could affect the accuracy of the findings.

D. Differing results between natural-habitat and lab studies of wild animals such as the Western scrub-jay are a strong indication that both of the studies had design flaws that affected the accuracy of their results.